Markov Decision Process (with reward)

$$\mathcal{M} = (\mathcal{S}, \mu_0, \mathcal{A}, P, R, \gamma)$$

$\mathcal{S}$: state space, set of all possible states

μ_0 : initial state distribution ; $\mu_0 \in \Delta(\mathcal{S})$

$\mathcal{A}$: action space, set of all possible actions

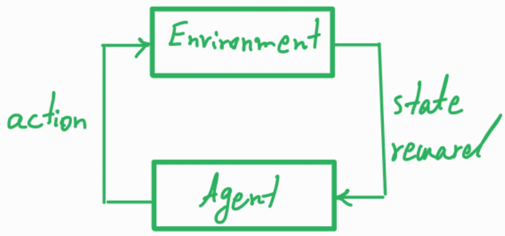
P : transition function ; $P : \mathcal{S} \times \mathcal{A} \longrightarrow \Delta(\mathcal{S})$

R : reward function ; $R : \mathcal{S} \times \mathcal{A} \longrightarrow \mathbb{R}$

γ : discount factor ; $\gamma \in [0, 1]$

other types of reward

- $R : \mathcal{S} \longrightarrow \mathbb{R}$
- $R : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \longrightarrow \mathbb{R}$
- $R : \mathcal{S} \times \mathcal{A} \longrightarrow \mathbb{R}^K$
- $R : \mathcal{S} \times \mathcal{A} \longrightarrow \Delta(\mathbb{R})$



Agent's interaction with the environment:

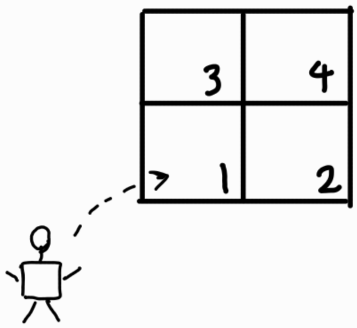
- The interaction starts at time step $t=0$ and state $s_0 \sim \mu_0$.

- At time step $t \in \mathbb{N}_0$ and state $s_t \in \mathcal{S}$, the agent takes action $A_t \in \mathcal{A}$, observes the next state $s_{t+1} \sim P(\cdot \mid s_t, A_t)$, and receives the immediate reward $R_t = R(s_t, A_t) \in \mathbb{R}$.

- The history of interaction at time t is called a trajectory $\tau_t \in \mathcal{T}$, where

$$\tau_t = (\underbrace{s_0, a_0, r_0}_{t=0}, \underbrace{s_1, a_1, r_1}_{t=1}, \dots, s_t, a_t, r_t)$$

Example:

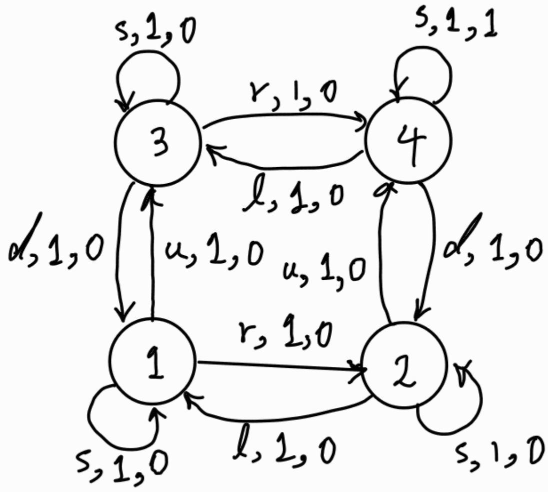


$S = \{1, 2, 3, 4\}$

$A = \{r, l, u, d, s\}$

$\mu_0 = [1, 0, 0, 0]$

MDP



edge label: (action, transition probability, reward)

sample trajectories

$(s_0 = 1, a_0 = u, r_0 = 0, s_1 = 3, a_1 = s, r_1 = 0, s_2 = 3 \ldots)$

$(s_0 = 1, a_0 = r, r_0 = 0, s_1 = 2, a_1 = u, r_1 = 0, s_2 = 4, a_2 = s, r_2 = 1, \ldots)$

Policy: A possibly randomized mapping from a trajectory to actions;

$$\pi : \underbrace{\mathcal{T}}_{\text{set of all possible trajectories}} \longrightarrow \Delta(A)$$

Until time H, there could be up to $|S|^H |A|^H$ trajectories.

- stationary policy: A policy where the action only depends on the current state;

$$\pi : S \longrightarrow \Delta(A)$$

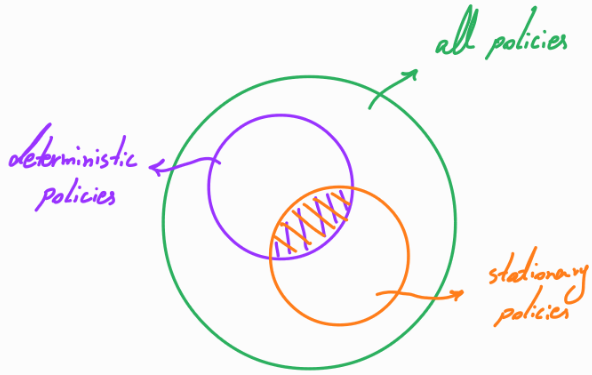
- Deterministic policy: A policy where the action is chosen deterministically;

$$\pi : \mathcal{T} \longrightarrow A$$

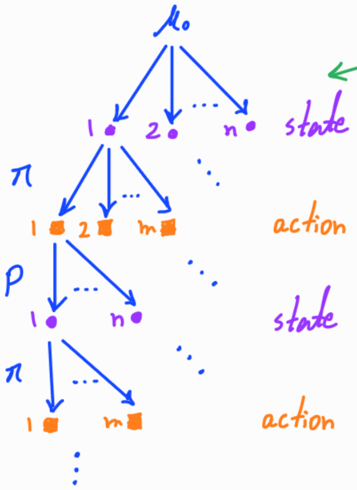
- Deterministic, stationary policy:

$$\pi : S \longrightarrow A$$

$|A|^{|S|}$ possible functions



Goal: The learning agent aims to find a policy π that maximizes the expected (discounted) cumulative reward



$$\pi^* \in \max_{\pi \in \Pi} \underbrace{\mathbb{E}_{\substack{S_0 \sim \mu_0 \\ A_t \sim \pi(\tau_t) \\ S_{t+1} \sim P(\cdot \mid S_t, A_t)}}}_{\text{average over all trajectories}} \Big[\overbrace{\sum_{t=0}^{T} \gamma^t R(S_t, A_t)}^{\text{return}} \Big]$$

for stochastic reward $\longrightarrow R(S_t, A_t) \sim \mathcal{R}(\cdot, \cdot)$

Average-reward setting

$$\lim_{T \to \infty} \mathbb{E}_{\substack{S_0 \sim \mu_0 \\ A_t \sim \pi(\tau_t) \\ S_{t+1} \sim P(\cdot \mid S_t, A_t)}} \Big[\frac{1}{T} \sum_{t=0}^{T-1} R(S_t, A_t) \Big]$$

$T \to T = \infty$ infinite horizon

$T \to T < \infty$ finite horizon

$\gamma \to \gamma = 1$ undiscounted

$\gamma \to 0 < \gamma < 1$ discounted

$\gamma \to \gamma = 0$ greedy / myopic

* We will start with the infinite-horizon discounted setting.

Value functions

- <u>(state) value function</u> of a policy π, $V^\pi : \mathcal{S} \longrightarrow \mathbb{R}$, is defined as

$$V^\pi(s) = \mathbb{E}_{\substack{A_t \sim \pi(\tau_t) \\ S_{t+1} \sim P(\cdot \mid S_t, A_t)}} \Big[\sum_{t=0}^{\infty} \gamma^t R(S_t, A_t) \;\Big|\; S_0 = s \Big] \qquad \forall s \in \mathcal{S}$$

- <u>(state-) action value function</u> of a policy π, $Q^\pi : \mathcal{S} \times \mathcal{A} \longrightarrow \mathbb{R}$ is defined as

$$Q^\pi(s,a) = \mathbb{E}_{\substack{A_t \sim \pi(\tau_t) \\ S_{t+1} \sim P(\cdot \mid S_t, A_t)}} \Big[\sum_{t=0}^{\infty} \gamma^t R(S_t, A_t) \;\Big|\; S_0 = s, A_0 = a \Big] \quad \forall s \in \mathcal{S}$$

$t \geq 1$

$$\underbrace{J(\pi)}_{\text{RL objective}} = \mathbb{E}_{\mu_0, \pi, P} \Big[\sum_{t=0}^{\infty} \gamma^t R(S_t, A_t) \Big] \longrightarrow J(\pi) = \sum_{s \in \mathcal{S}} \mu_0(s) V^\pi(s) = \mathbb{E}_{\mu_0}[V^\pi(s)]$$

$\longrightarrow$ if $\mu_0 = [0 \; 0 \; \dots \; 0 \; \underset{S=s}{1} \; 0 \dots 0] \Longrightarrow V^\pi(s) = J(\pi)$

Bellman (consistency) Equations:

Let π be a stationary policy. The value functions under π, i.e., V^π and Q^π, satisfy the following equations:

$$V^\pi(s) = \mathbb{E}_{a\sim\pi(s)}\Big[\overbrace{R(s,a) + \gamma\, \mathbb{E}_{s'\sim P(\cdot|s,a)}\big[V^\pi(s')\big]}^{Q^\pi(s,a)}\Big]$$

$$Q^\pi(s,a) = R(s,a) + \gamma\, \mathbb{E}_{s'\sim P(\cdot|s,a)}\big[V^\pi(s')\big]$$

Proof:

$$Q^\pi(s,a) = \mathbb{E}_{\pi,P}\Big[\sum_{t=0}^{\infty} \gamma^t R(S_t,A_t) \,\Big|\, S_0=s, A_0=a\Big]$$

$$= \mathbb{E}_{\pi,P}\Big[R(S_0,A_0) + \sum_{t=1}^{\infty} \gamma^t R(S_t,A_t) \,\Big|\, S_0=s, A_0=a\Big]$$

$$= R(s,a) + \mathbb{E}_{\pi,P}\Big[\sum_{t=1}^{\infty} \gamma^t R(S_t,A_t) \,\Big|\, S_0=s, A_0=a\Big]$$

$$\begin{aligned}\mathbb{P}[X|Y] &= \sum_{z\in Z} \mathbb{P}[X,Z|Y] \\ &= \sum_{z\in Z} \mathbb{P}[X|Y,Z]\,\mathbb{P}[Z|Y]\end{aligned}$$

$$= R(s,a) + \sum_{s'\in S}\Big(\mathbb{E}_{\pi,P}\Big[\sum_{t=1}^{\infty} \gamma^t R(S_t,A_t) \,\Big|\, S_0=s, A_0=a, S_1=s'\Big]\ \mathbb{P}\big[S_1=s' \,|\, S_0=s, A_0=a\big]\Big)$$

$$= R(s,a) + \sum_{s'\in S}\Big(\gamma\, \mathbb{E}_{\pi,P}\Big[\underbrace{\sum_{t=1}^{\infty} \gamma^{t-1} R(S_t,A_t)}_{R(S_1,A_1)+\gamma R(S_2,A_2)+\gamma^2 R(S_3,A_3)+\cdots} \,\Big|\, S_1=s'\Big] P(s'|s,a)\Big)$$

$$= R(s,a) + \gamma \sum_{s'\in S}\big(V^\pi(s')\, P(s'|s,a)\big)$$

$$= R(s,a) + \gamma\, \mathbb{E}_{s'\sim P(\cdot|s,a)}\big[V^\pi(s')\big]$$

The first equation can also be derived from this.

Theorem [sufficiency of search over deterministic, stationary policies]:

There exists an optimal, deterministic, stationary policy π^*, i.e.,

$$\exists \pi^* : \mathcal{S} \longrightarrow \mathcal{A} \quad \text{s.t.} \quad V^{\pi^*}(s) \geq V^{\pi}(s) \quad \forall s \in \mathcal{S}, \forall \pi \in \Pi .$$

Proof (sketch): Consider a deterministic, stationary policy $\tilde{\pi}$ as

$$\tilde{\pi}(s) = \sup_{a \in \mathcal{A}} \mathbb{E}_{s' \sim P(\cdot|s,a)} \left[R(s,a) + \gamma V^*(s') \mid S_0 = s, A_0 = a \right] \quad \forall s \in \mathcal{S}$$

$$V^*(s) = \sup_{\pi \in \Pi} V^{\pi}(s) \quad \forall s \in \mathcal{S}$$

To show that $\tilde{\pi}$ is an optimal policy, we need to show that

$$V^{\tilde{\pi}}(s) = V^*(s) \quad \forall s \in \mathcal{S}$$

- $V^{\tilde{\pi}}(s) \leq V^*(s)$
- $V^{\tilde{\pi}}(s) \geq V^*(s)$

$$\begin{aligned} V^*(s_0) &\leq \mathbb{E}_{\tilde{\pi}, s_1} \left[R(s_0, A_0) + \gamma V^*(s_1) \right] \\ &\leq \mathbb{E}_{\tilde{\pi}, s_1, s_2} \left[R(s_0, A_0) + \gamma R(s_1, A_1) + \gamma^2 V^*(s_2) \right] \\ &\;\;\vdots \\ &\leq V^{\tilde{\pi}}(s_0) \end{aligned}$$

if $R(\cdot,\cdot)$ finite $\rightarrow V^*(\cdot)$ finite

Proof: Let $V^*(s) = \sup_{\pi \in \Pi} V^\pi(s) \quad \forall s \in S$

$\pi \in \Pi$ a general policy, i.e.,

$$\pi(A_t = a \mid \overbrace{S_0 = s_0, A_0 = a_0, R_0 = r_0, \dots, S_t = s_t}^{\tau_t})$$

is the probability of taking action a at time t based on the history.

step 1

> Intuition: The maximum future discounted value from $t=1$ onwards is independent of S_0, A_0, R_0 if S_1 is known.

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We want to show that

$$\sup_{\pi \in \Pi} \mathbb{E}_{\pi, P}\left[\sum_{t=1}^{\infty} \gamma^t R(S_t, A_t) \,\middle|\, (S_0, A_0, R_0, S_1) = (s, a, r, s')\right] = \gamma V^*(s')$$

- Let $\pi_{(s,a,r)}$ be an offset policy for π in the following sense

$$\pi_{(s,a,r)}(A_t = a \mid \underbrace{S_0 = s_0, A_0 = a_0, R_0 = r_0, \dots, S_t = s_t}_{\tau})$$

$$= \pi(A_{t+1} = a \mid \underbrace{S_0 = s, A_0 = a, R_0 = r, S_1 = s_0, A_1 = a_0, R_1 = r_0, \dots, S_{t+1} = s_t}_{(s,a,r,\tau)})$$

> Intuition: $\pi_{(s,a,r)}$ chooses actions for τ with the same distribution that π chooses actions for (s,a,r,τ).

$$\mathbb{E}_{\pi,P}\left[\sum_{t=1}^{\infty}\gamma^t R(S_t,A_t) \,\middle|\, (S_0,A_0,R_0,S_1)=(s,a,r,s')\right]$$

Markov property + change of variable for time index + definition of $\pi_{(s,a,r)}$

$$= \gamma\, \mathbb{E}_{\pi_{(s,a,r)},P}\left[\sum_{t=0}^{\infty}\gamma^t R(S_t,A_t) \,\middle|\, S_0=s'\right]$$

definition of value function

$$= \gamma\, V^{\pi_{(s,a,r)}}(s')$$

$\xRightarrow{\text{taking sup}}$

$$\sup_{\pi\in\Pi}\mathbb{E}_{\pi,P}\left[\sum_{t=1}^{\infty}\gamma^t R(S_t,A_t) \,\middle|\, (S_0,A_0,R_0,S_1)=(s,a,r,s')\right]$$

correspondence between $\pi_{(s,a,r)}$ and π

$$= \gamma \sup_{\pi\in\Pi} V^{\pi_{(s,a,r)}}(s') = \gamma \sup_{\pi\in\Pi} V^{\pi}(s') = \gamma\, V^*(s') \qquad (1)$$

definition of optimal value function

step 2

Consider the following deterministic, stationary policy

$$\tilde{\pi}(s) = \underbrace{\sup_{a\in\mathcal{A}}}_{\text{deterministic action}} \underbrace{\mathbb{E}_{S_1\sim P(\cdot|s,a)}\left[R(s,a)+\gamma V^*(S_1) \,\middle|\, (S_0,A_0)=(s,a)\right]}_{\text{depends on the current state, not the rest of the trajectory}} \quad \forall s\in\mathcal{S}$$

We want to show that $\tilde{\pi}$ is an optimal policy, i.e.,

$$V^{\tilde{\pi}}(s) = V^*(s) \quad \forall s\in\mathcal{S} \qquad (2)$$

$$v^*(s_0) = \sup_{\pi \in \Pi} v^\pi(s_0) = \sup_{\pi \in \Pi} \mathbb{E}_{\pi, p}\left[\sum_{t=0}^{\infty} \gamma^t R(S_t, A_t)\right]$$

split the sum

$$= \sup_{\pi \in \Pi} \mathbb{E}_{\pi, p}\left[R(s_0, A_0) + \sum_{t=1}^{\infty} \gamma^t R(S_t, A_t)\right]$$

law of total expectation

$$= \sup_{\pi \in \Pi} \mathbb{E}_{\underline{A_0, S_1}}\left[R(s_0, A_0) + \mathbb{E}_{\underline{\pi, p}}\left[\sum_{t=1}^{\infty} \gamma^t R(S_t, A_t) \,\middle|\, (S_0, A_0, R_0, S_1) = (s_0, a_0, r_0, s_1)\right]\right]$$

$$\underline{\leq} \sup_{\pi \in \Pi} \mathbb{E}_{A_0, S_1}\left[R(s_0, A_0) + \sup_{\underline{\pi' \in \Pi}} \mathbb{E}\left[\sum_{t=1}^{\infty} \gamma^t R(S_t, A_t) \,\middle|\, (S_0, A_0, R_0, S_1) = (s_0, a_0, r_0, s_1)\right]\right]$$

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$$= \sup_{\pi \in \Pi} \mathbb{E}_{A_0, S_1}\left[R(s_0, A_0) + \gamma v^*(s_1)\right]$$

the only variation in the operand of sup is due to A_0

$$= \sup_{a_0 \in \mathcal{A}} \mathbb{E}_{\underline{S_1}}\left[R(s_0, a_0) + \gamma v^*(s_1)\right]$$

definition of $\tilde{\pi}$

$$= \mathbb{E}_{\tilde{\pi}, S_1}\left[R(s_0, a_0) + \gamma v^*(s_1)\right]$$

Applying the same argument repeatedly in a recursive way $\longrightarrow$

$$v^*(s_0) \leq \mathbb{E}_{\underline{\tilde{\pi}, S_1}}\left[R(s_0, A_0) + \gamma v^*(s_1)\right]$$

unrolling of the sum

$$\left.\begin{aligned} &\leq \mathbb{E}_{\underline{\tilde{\pi}, S_1, S_2}}\left[R(s_0, A_0) + \gamma R(S_1, A_1) + \gamma^2 v^*(S_2)\right] \\ &\vdots \\ &\leq v^{\tilde{\pi}}(s_0) \end{aligned}\right\}$$

On the other hand,

$$v^{\tilde{\pi}}(s_0) \leq \sup_{\pi \in \Pi} v^{\pi}(s_0) = v^*(s_0) \qquad \Longrightarrow v^{\tilde{\pi}}(s_0) = v^*(s_0)$$

Note that this is true for all $s_0 \in \mathcal{S}$ $\Longrightarrow$

$$v^{\tilde{\pi}}(s) = v^*(s) \qquad \forall s \in \mathcal{S} \qquad (2)$$

Corollary: Consider a deterministic, stationary policy $\tilde{\pi}$ as

$$\tilde{\pi}(s) = \sup_{a \in \mathcal{A}} \mathbb{E}_{s' \sim P(\cdot|s,a)} \left[R(s,a) + \gamma v^*(s') \,\middle|\, S_0 = s, A_0 = a \right] \quad \forall s \in \mathcal{S}$$

The following holds:

$$Q^{\tilde{\pi}}(s,a) = Q^*(s,a) \qquad \forall s \in \mathcal{S}, \forall a \in \mathcal{A},$$

where $\quad Q^*(s,a) = \sup_{\pi \in \Pi} Q^{\pi}(s,a) \qquad \forall s \in \mathcal{S}, \forall a \in \mathcal{A}.$